```
#include "HX711.h"

HX711 scale1;
HX711 scale2;
HX711 scale3;
HX711 scale4;

int SCK1 = 11;
int DT1 = 10;

int SCK2 = 9;
int DT2 = 8;

int SCK3 = 7;
int DT3 = 6;

int SCK4 = 5;
int DT4 = 4;

int tare_button = 3;

int last_state = HIGH;

int trials = 2;

bool do_tare = false;

const int N = 20;

float buf1[N]={0};
float buf2[N]={0};
float buf3[N]={0};
float buf4[N]={0};
int avgInd = 0;


void tareAll(HX711 &s1,HX711 &s2,HX711 &s3,HX711 &s4){
  delay(500);
  s1.tare();
  s2.tare();
```

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  s3.tare();
  s4.tare();
}

void extremeTare(HX711 &scale1,HX711 &scale2,HX711 &scale3,HX711 &scale4){
  delay(500);
  tareAll(scale1,scale2,scale3,scale4);
  int iter = 50;
  int tareTrials = 1;
  int threshold = 100;
  int maxAttempts = 10;
  float scale1Avg=0;
  float scale2Avg=0;
  float scale3Avg=0;
  float scale4Avg=0;

  for(int i=0;i<maxAttempts;i++){
    scale1Avg=0;
    scale2Avg=0;
    scale3Avg=0;
    scale4Avg=0;
    tareAll(scale1,scale2,scale3,scale4);
    for(int i=0;i<iter;i++){
      scale1Avg += scale1.get_units(tareTrials);
      scale2Avg += scale2.get_units(tareTrials);
      scale3Avg += scale3.get_units(tareTrials);
      scale4Avg += scale4.get_units(tareTrials);
      delay(100);
    }
    scale1Avg /= iter;
    scale2Avg /= iter;
    scale3Avg /= iter;
    scale4Avg /= iter;
    if (abs(scale1Avg) <= threshold || abs(scale2Avg) <= threshold ||
abs(scale3Avg) <= threshold || abs(scale4Avg) <= threshold){break;}

  }
  for(int i = 0; i < N; i++){
    buf1[i] = scale1.get_units(1);
    buf2[i] = scale2.get_units(1);
```

```
    buf3[i] = scale3.get_units(1);
    buf4[i] = scale4.get_units(1);
    delay(100);
  }
  avgInd = 0;
}

float getAvg(float buf[], int N){
  float sum = 0;
  for(int i=0;i<N;i++){
    sum += buf[i];
  }
  return sum / N;
}

void setup() {
  // put your setup code here, to run once:
  Serial.begin(115200);
  scale1.begin(DT1,SCK1);
  scale2.begin(DT2,SCK2);
  scale3.begin(DT3,SCK3);
  scale4.begin(DT4,SCK4);

  scale1.set_scale(1.0f);
  scale2.set_scale(1.0f);
  scale3.set_scale(1.0f);
  scale4.set_scale(1.0f);

  pinMode(tare_button,INPUT_PULLUP);
  delay(500);
  scale1.tare();
  scale2.tare();
  scale3.tare();
  scale4.tare();
  delay(2000);
  extremeTare(scale1,scale2,scale3,scale4);
  //Serial.println("NOW");
  delay(3000);

}
```

```
void loop() {

  if (scale1.is_ready() && scale2.is_ready() && scale3.is_ready() &&
scale4.is_ready()) {
    buf1[avgInd] = scale1.get_units(trials);
    buf2[avgInd] = scale2.get_units(trials);
    buf3[avgInd] = scale3.get_units(trials);
    buf4[avgInd] = scale4.get_units(trials);

    avgInd = (avgInd + 1) % N;
    Serial.print(getAvg(buf1, N));
    Serial.print(",");
    Serial.print(getAvg(buf2, N));
    Serial.print(",");
    Serial.print(getAvg(buf3, N));
    Serial.print(",");
    Serial.println(getAvg(buf4, N));
  }

  float avg1 = 0, avg2 = 0, avg3 = 0, avg4 = 0;

  if (digitalRead(tare_button) == LOW && last_state == HIGH){
    do_tare = true;
  }
  last_state = digitalRead(tare_button);


  if (do_tare == true){
    delay(1500);
    extremeTare(scale1,scale2,scale3,scale4);
    do_tare = false;
    //Serial.println("TARED!!");
    delay(500);
  }

}
```